SMITHA SOMAN

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SUMMARY

M.S. Artificial Intelligence student (December 2027) with 3.5 years of production embedded C engineering for automotive controllers at Bosch and Navistar. Combines real-time systems, CAN diagnostics, and hardware/software debugging with graduate work in ML robustness, model evaluation, and quantized inference. Interested in robotics, autonomy, and reliable AI deployment.

EDUCATION

M.S. Artificial Intelligence, San Jose State University Jan 2026 – Expected Dec 2027
B.E. Electronics & Communication, Amrita Vishwa Vidyapeetham, India
Relevant coursework: Deep Learning, Machine Learning, AI and Data Engineering, Database Systems, Data Structures in C++

TECHNICAL SKILLS

Languages: Python, C, C++, SQL, Shell, CAPL
AI systems: PyTorch, model evaluation, post-training quantization, CPU latency benchmarking, uncertainty estimation, bootstrap analysis, scikit-learn
Robotics & cloud: LIBERO-Plus, MuJoCo simulation, LeRobot, Google Cloud GPUs
Embedded: Real-time OS, ECU software, CAN/CANoe/CANalyzer, diagnostics, hardware/software integration
Tools: Git, pytest, Jupyter, PostgreSQL, SQLAlchemy

TECHNICAL EXPERIENCE

Senior Product Development Engineer | Navistar Inc., Lisle, IL Aug 2010 – Oct 2012
- Developed embedded application software for the Electronic System Controller (ESC) on commercial vehicle platforms.
- Debugged and customized kernel-level C code in a real-time OS to improve real-time performance and reduce control-loop latency.
- Designed Driver Information Display functionality; used CANoe and CAPL for CAN simulation and testing.

Software Engineer | Robert Bosch Engineering & Business Solutions, India Aug 2008 – Nov 2009
- Developed production embedded C for diesel engine control units in vehicle motion demand, communication, and device encapsulation subsystems for Daimler, Chrysler, and Toyota programs.
- Owned requirements analysis, design, implementation, code review, testing, and ECU integration.
- Built CAN-based dashboard display, signal configuration, freeze-frame diagnostics, and stall-detection functionality; validated traffic using CANalyzer and CANvas.

SELECTED AI & ROBOTICS PROJECTS

Quantized Vision-Language-Action Robustness | CMPE 252 Fall 2026 (in progress)
- Studying whether 4-bit quantization of the vision encoder in π0, a 3.3B-parameter VLA robot policy, stays near-lossless under lighting, background, and layout shifts, using matched closed-loop LIBERO-Plus episodes and a difference-in-differences robustness metric.
- Own the LIBERO-Plus integration (benchmark runner, shift sampling, base-state mapping), the visual-shift experiments, and the primary analysis with task-clustered bootstrap intervals; set up the team's GPU infrastructure on Google Cloud. Results pending.

Audio Deepfake Detection Under Distribution Shift | CMPE 258 Fall 2026 (in progress)
- Own the evaluation harness for an ASVspoof 5 shift study: protocol parsing, paired codec and adversarial slices, speaker-level bootstrap intervals, and every results table and figure.
- Harness reproduces the official ASVspoof 5 AASIST baseline (15.20% dev EER vs. 15.2% published); running falsification controls and DFADD cross-dataset evaluation, with INT8 CPU latency benchmarking planned.

Resource-Efficient Uncertainty for TinyML (QUTE) | CMPE 257 Fall 2026 (in progress)
- Re-implementing an ICML 2025 single-pass ensemble with no public code; own the base CNN and shared prediction interface the team builds on (13 unit tests), the early-exit branches and two output heads, the weight-transfer training loop, and its ablation.
- Comparing against softmax confidence, MC Dropout, and deep ensembles on calibration, label-free accuracy-drop detection across 15 MNIST-C corruptions, and parameter cost.

Text-to-SQL Pipeline | Independent Summer 2026
- Built a modular Python LLM-to-SQL pipeline with schema introspection, pluggable model backends, and AST-based query validation.
- Enforced SELECT-only, schema-checked execution using sqlglot allow-lists; added 29 pytest tests runnable without API keys.

ADDITIONAL EXPERIENCE

Program Manager | PlayCV, Castro Valley, CA Aug 2022 – Aug 2025
- Led operations across five locations and 90+ employees, including hiring, training, systems rollout, and process improvement.

Founder & Arts Educator | Independent Arts Education Studio Nov 2012 – Aug 2022
- Founded and operated an education business, overseeing instruction, customer relationships, and operations.